Camilo Artigas

(647) 836-0531 | cartigas@uwaterloo.ca

Dual US & Canadian Citizen

Website | LinkedIn | GitHub

Experience

Electrical Engineering Intern - Powertrain / Avionics **Sep. 2026 – Dec. 2026**

Zipline *South San Francisco, CA*

- Leading redesign of aircraft servo controller, implementing and testing new inverter, cascaded buck converters and current sensors while owning schematic capture and layout in Altium.
- Owned electrical implementation and instrumentation of avionics connector vibration test leg by designing schematics and layout of 6 layer FPGA (Spartan Ultrascale) Bit Error Rate Tester and Camera/FAKRA PCB in Altium.
- Integrated current sensors into existing 8-layer motor control PCB in Altium, owning schematics, layout, and bring-up to test efficacy against shunt/op-amp alternative, successfully revving PCB.
- Designed flex PCB in Altium to provide telemetry between winch and hall effect sensor for 100+ EOL testers.
- Designed layout of 2 HIL FPGA interposer PCBs, routing PCIexpress traces and integrating peripherals within 6 layers.
- Designed breakout PCB in Altium for winch motor controller EOL testing. Routed and impedance matched differential pairs, cutting setup time by 50%

Mechatronics Engineering Intern - Powertrain **Jan. 2026 – May 2026**

Zipline *South San Francisco, CA*

- Enabled full aircraft, droid & dock accelerated testing by retrofitting a shipping container into an environmental chamber, integrating an Arduino Opta and 8 RS-485 sensors to actuate heaters and misters via relay closed loop control.
- Developed multi-threaded C++ firmware for real-time control of chamber via RS-485 and a UART Python dashboard for data visualization; tuning duty cycles and tolerances to maintain 65 °C and 85% RH for 30 days.
- Owned an EV3 Battery test campaign by designing 2 PCBs in Altium and a 2000 Hz first-mode vibration fixture in NX.
- Designed full aircraft vibration fixture in NX for Busbar component testing, leveraging Modal Analysis to push first modal frequency beyond 500 Hz to enable launch-critical validation.
- Designed a Motor Rotor Tester in NX to validate magnet epoxy strength at 30,000+ RPM via a 1:5 belt drive reduction.

Hardware Engineering Intern - Sensors **May 2025 – Aug. 2025**

Kodiak Robotics *Mountain View, CA*

- Designed and brought up a modular sensor cleaning test bench, integrating an in-house PCB with C++ firmware and Python test scripts to enable repeatable, plug-and-play validation across multiple sensors, reducing setup time by 70%.
- Designed and prototyped a liquid and air sensor cleaning system with 12 nozzles, optimizing pressure and angle for LIDAR, WFOV, and NFOV cameras — ensuring reliable perception for fully autonomous trucks.
- Designed a sheet metal enclosure in SolidWorks to house pneumatics for the air cleaning system, applying DFMA principles to reduce on-road troubleshooting time by 30% and personally mounting the full assembly onto the truck.

Mechatronics Engineering Intern **Sep. 2024 – Dec. 2024**

Sheartak Tools Ltd. *Waterloo, ON*

- 3D modelled 10+ spiral cutterheads in SolidWorks, drafting technical drawings using GD&T to specify critical tolerances.
- Developed Python scripts with OpenPyXL and NumPy, automating the migration of 500+ products, saving 30+ hours.

Student Design Teams

Hardware Director & Founder **Dec. 2025 – Present**

Waterloo Automation Collective *Waterloo, ON*

- Leading all hardware development of UWaterloo's first injection molder. Running meetings, design reviews and mentoring 15+ students.

Hardware Lead & Designer **Aug. 2024 – Dec. 2025**

Waterloo Aerial Robotics Group *Waterloo, ON*

- Co-led the mechanical team ahead of the 2026 competition, owning and managing the Task 1 Payload within the Hexacopter assembly. Leading meetings, design reviews and mentoring 15+ students.

Skills

EE: Altium, LTSpice, Bring-up, Prototyping & Debugging, Soldering & Harnessing, Oscilloscope, Power Electronics, Sensors
SWE / FW: Python, C++, Arduino, STM32, I2C, SPI, UART, RS-485/Modbus, MATLAB, Git, Linux, NumPy, Matplotlib
ME: Siemens NX, SolidWorks, GD&T, DFMA, Structural/Modal FEA, FDM/SLA Printing, Mill, Lathe, CNC, Sheet Metal

Education

University of Waterloo **Sep. 2023 – Apr. 2028**

Candidate for a Bachelor of Applied Science in Mechatronics Engineering *Waterloo, ON*

Relevant Coursework: Electronic Circuits, Power Electronics, Sensors & Instrumentation, Embedded Systems, Digital Logic

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Hardware @ Zipline | *Siemens NX, Altium, Systems Integration, Test Engineering, Modal Analysis*

Owned multiple test legs for the EV3 Drone over my first stint at Zipline. Bringing up testers such as:

- Environmental chamber capable of maintaining 65 °C and 85% RH for 30 days with two drones and docks inside.
- Test motor control PCB and 2 battery testing PCBs.
- 30,000 RPM Motor Rotor tester to validate epoxy strength.
- A full drone vibration fixture with a first mode beyond 500 Hz.



EV3 Drone Taking Off



Docked EV3 Drone

Hardware @ Kodiak | *SolidWorks, Python, Systems Integration, Test Engineering, Pneumatics*

Designed and prototyped a liquid and air sensor cleaning system optimizing pressure and angle for a LiDAR and WFOV/NFOV cameras. Actuated it by bringing up a modular sensor cleaning test bench, integrating an in-house PCB with Python and C++ automation scripts. Validated full system by capturing synchronized images and LiDAR PCAPs during operation.



Kodiak Commercial Sensor Pod Assembly



Sensors in cleaning system: WFOV, NFOV Cameras and OT-128 LiDAR

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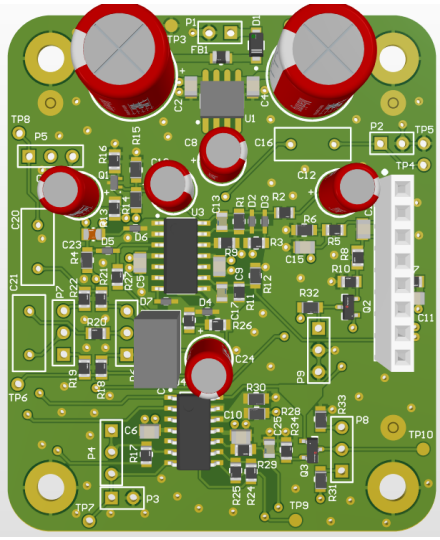
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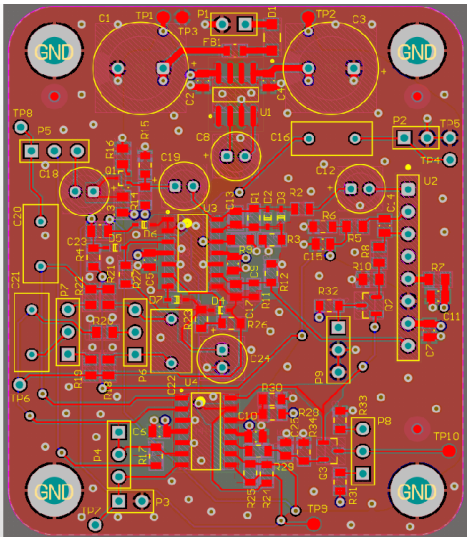
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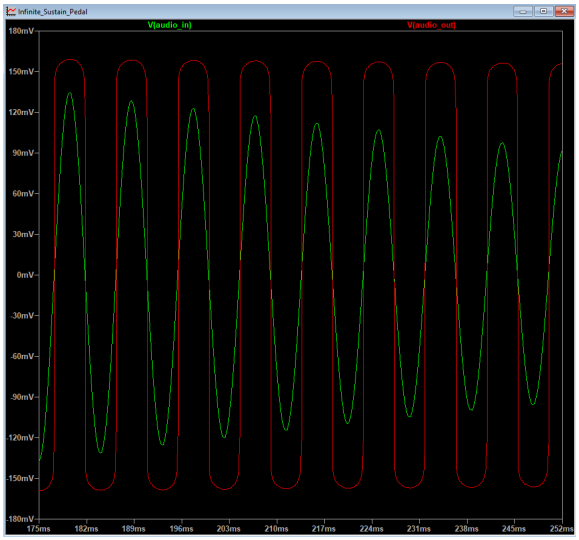
Infinite Sustain Guitar Pedal | *Altium, LT Spice, Circuit Design, Signal Processing*

Designed 4 layer infinite sustain pedal PCB in Altium. Designed and validated circuit from ground up in LT Spice using transient analysis. Implemented 8 op-amp stages, charge pump, voltage controlled amplifier and baxendall filter.





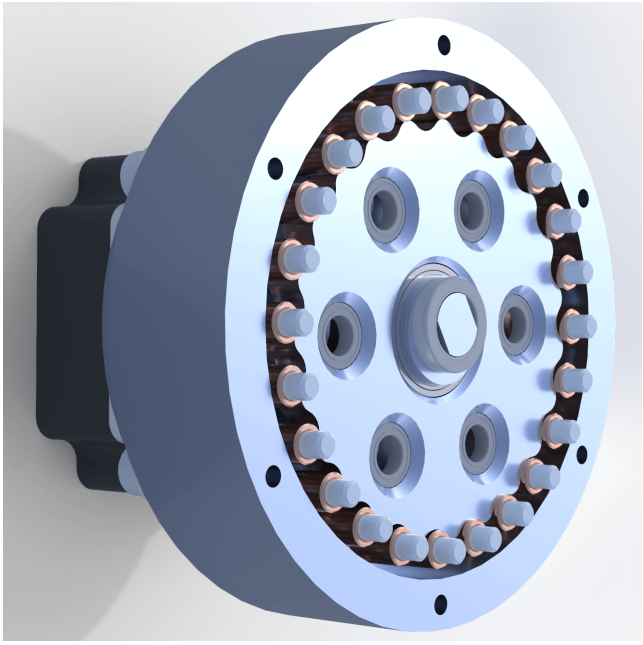
Altium PCB/Layout



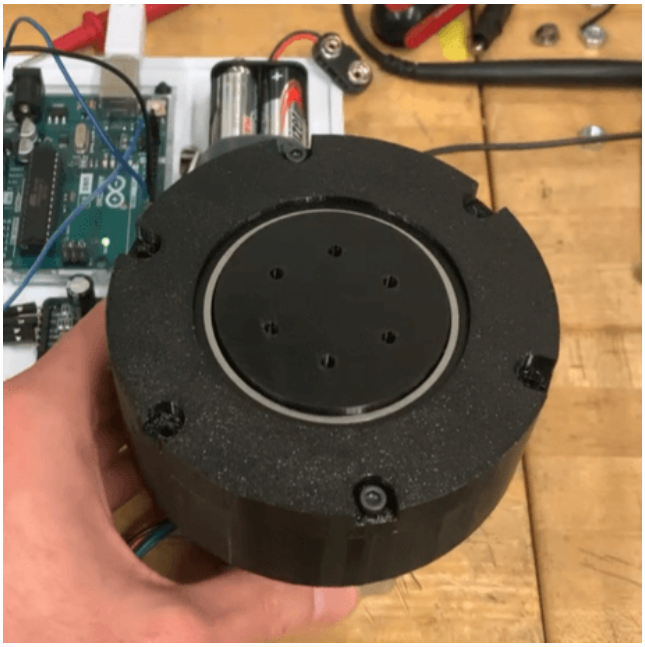
LT Spice Simulation
[Input → Green, Output → Red]

Cycloidal Actuator | *SolidWorks, 3D Printing, DFMA, Python, Stepper Motors*

Designed a 130+ part assembly with gear ratio of 23:1 using a cycloidal speed reducer. Developed a Python script for instant visualization of the cycloid, reducing design time by 50%. 3D printed a backlash-free prototype. (Code)



SolidWorks Model



Functional Prototype → WATCH HERE

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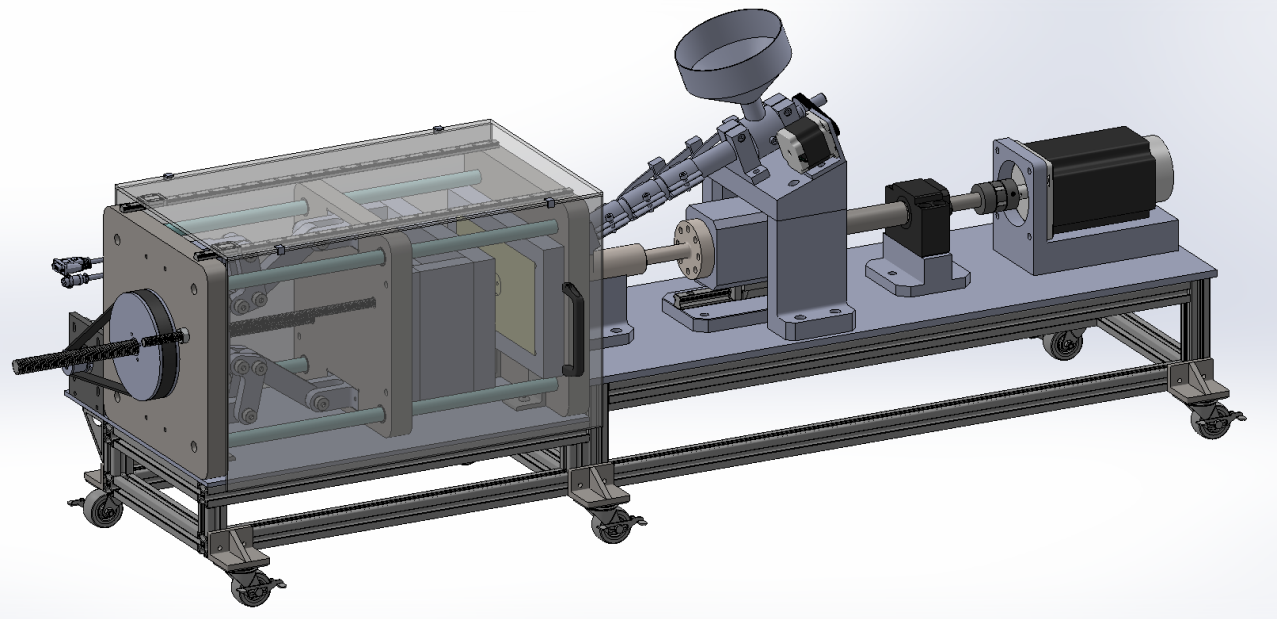
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Injection Molder | *SolidWorks, DFMA, Systems Integration, Stepper Motors*

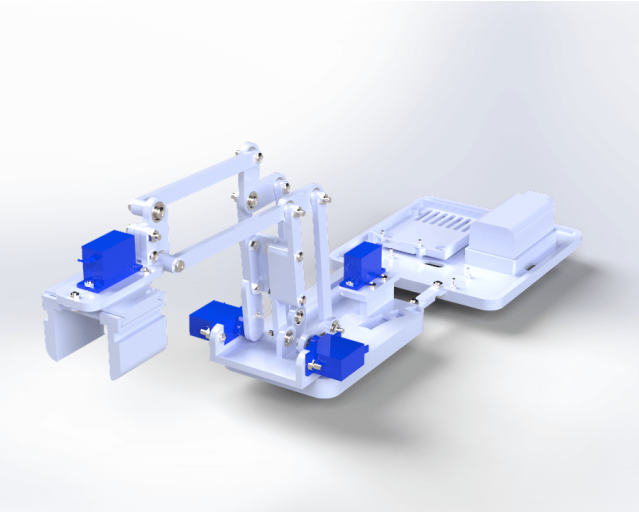
Founded and currently leading a 15+ member team building an injection molder at UWaterloo. Owned the injection unit and electronics unit assemblies. Designed it to hit 100+ MPa injection pressure and 3 in^3 shot volume.



SolidWorks Assembly

3-DOF Robotic Arm | *SolidWorks, 3D Printing, DFMA, Python, C++, UART, Servo Motors*

Designed a robotic arm and derived its inverse kinematics, achieving movement along the x, y, and z axes using servo motors. Established serial communication between a Python GUI and Arduino IDE for precise control. (Code)



SolidWorks Model



Robotic Arm Controlled with GUI → WATCH HERE